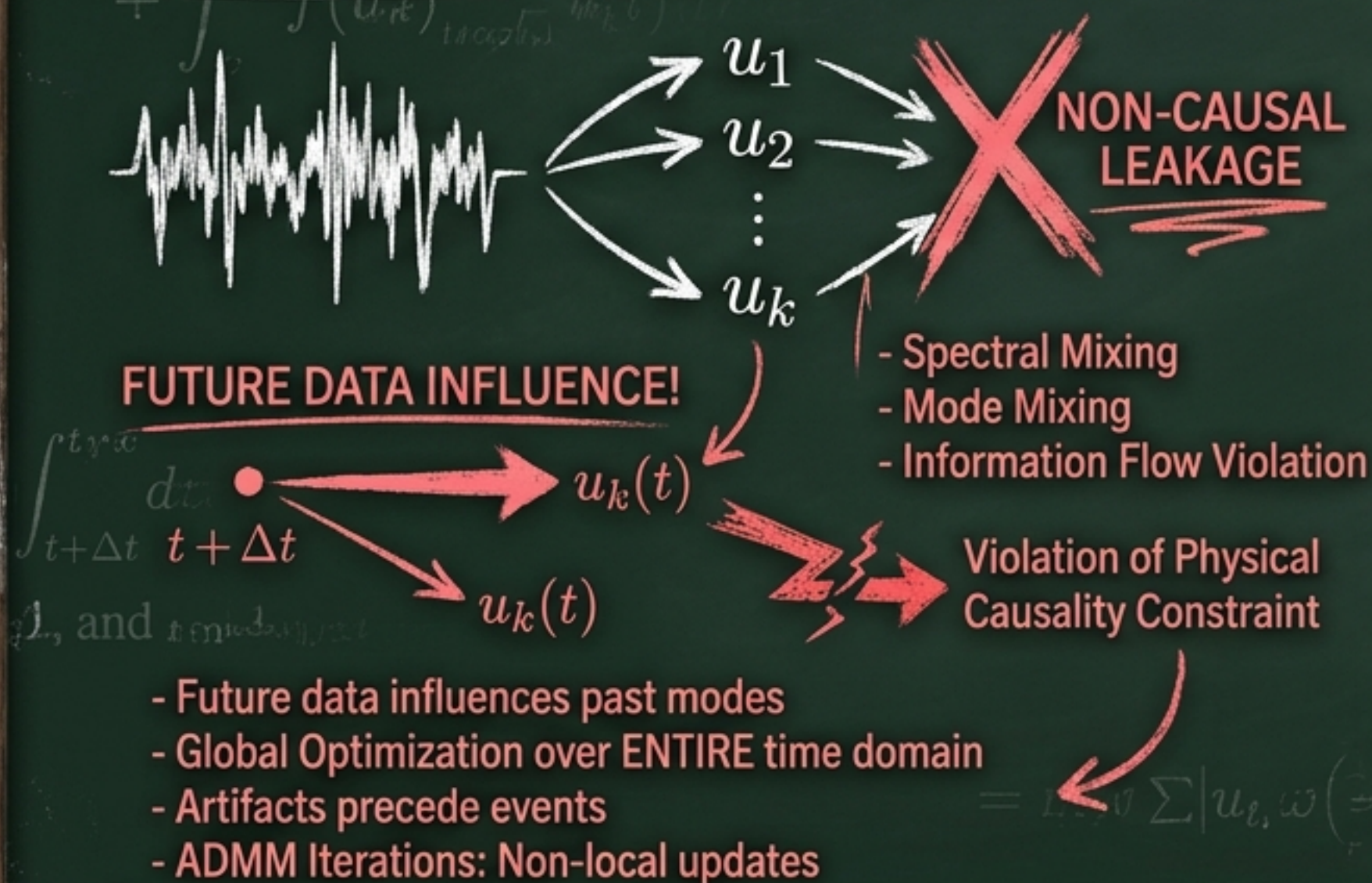


The VMD Causality Paradox in Time Series SDEs

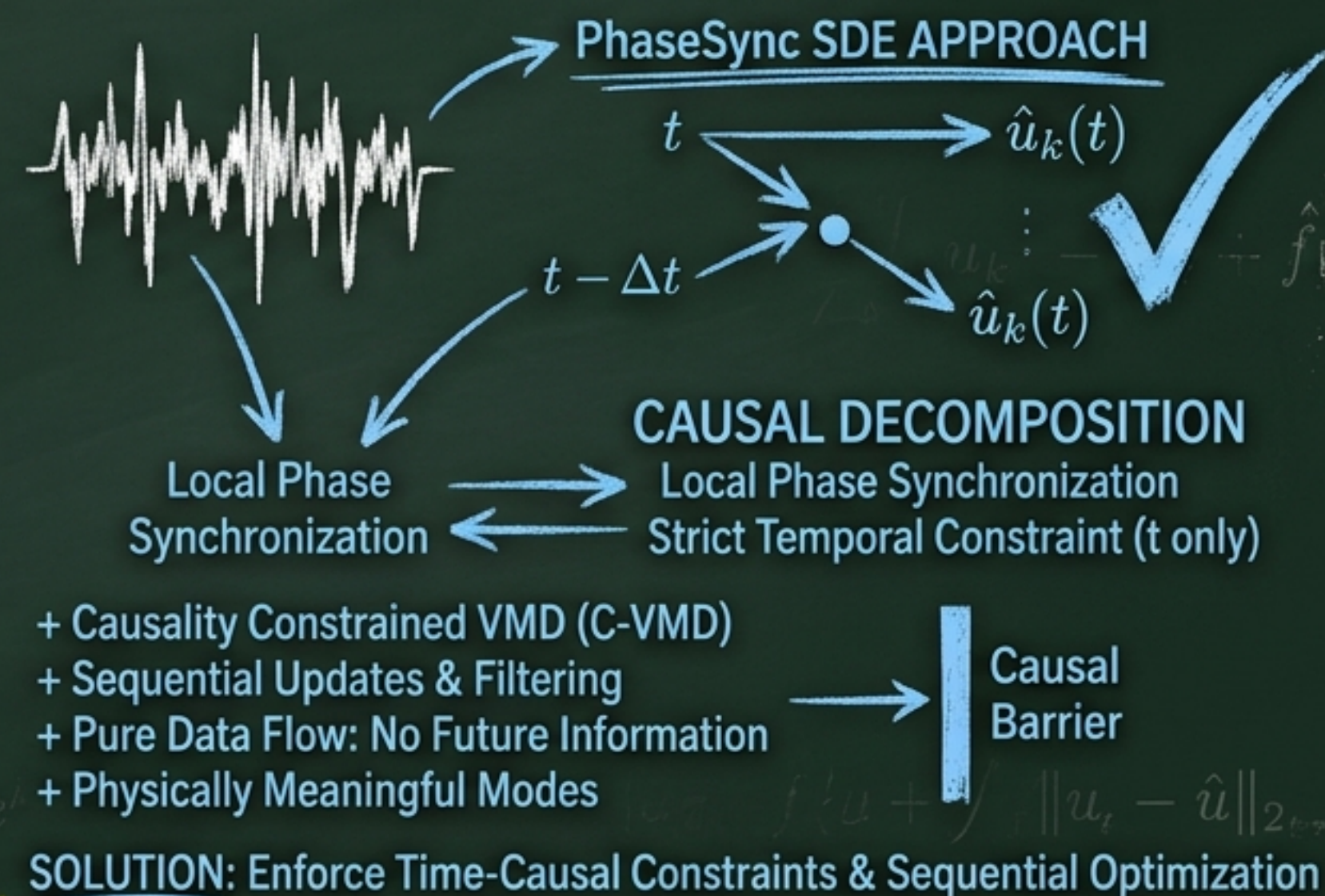
Solving Non-Causal Leakage & Global Optimization in PhaseSync SDE

THE VMD CAUSALITY PARADOX



HAZARD: Inaccurate Forecasting & Invalid Physical Interpretation!

SOLVING NON-CAUSAL LEAKAGE: PhaseSync SDE



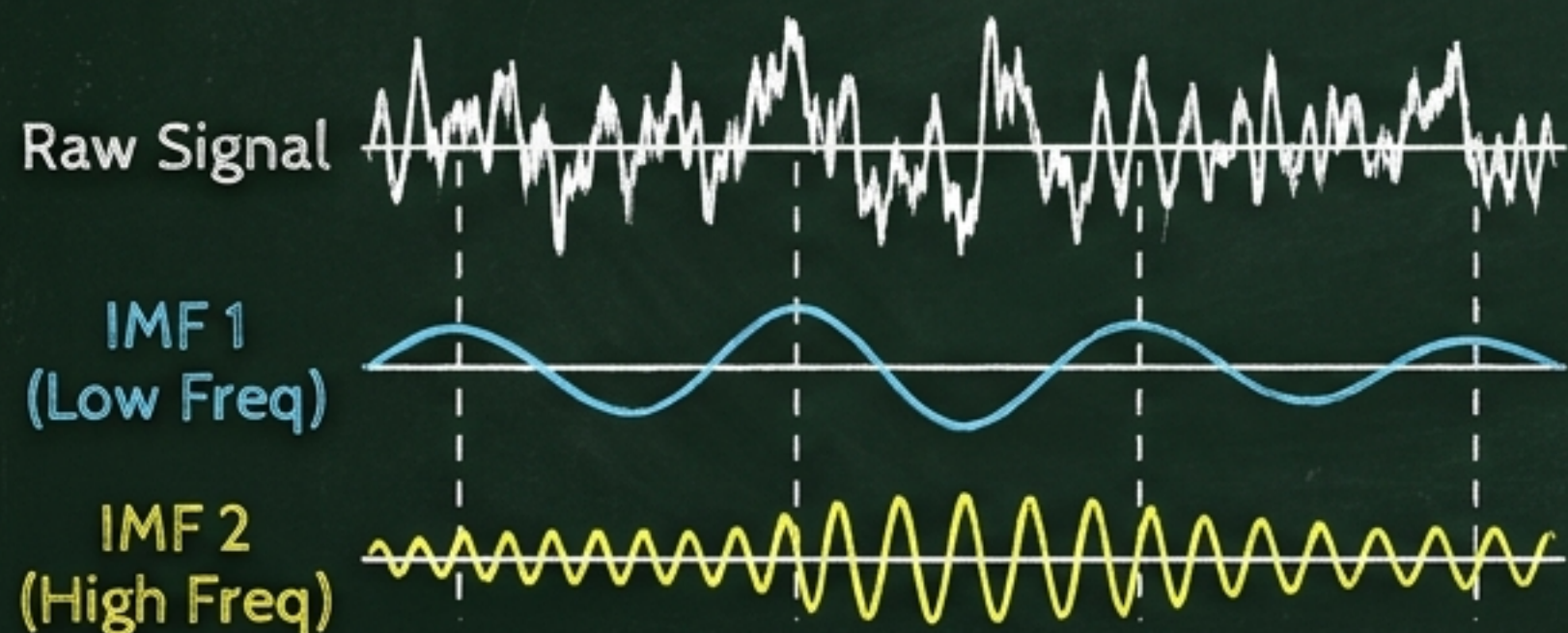
★ **Key Takeaway:** Shift from **GLOBAL** to **CAUSAL FRAMEWORK**

Does VMD Preserve Data Order?

✓ YES: Time Alignment

VMD outputs match the exact length and time index of the input.

- $f(t)$ at $t=100$ aligns perfectly with $\text{IMF}(t)$ at $t=100$.
- Time-frequency locality is beautifully preserved.
- It does NOT shuffle or scramble chronological order.



✗ NO: Strict Causality

VMD calculations for a past point use data from the future.

- To calculate IMF at $t=100$, VMD looks ahead at $t=101 \dots 240$.
- It operates as a bi-directional, non-causal filter.
- Future market crashes leak backward into past mode extractions.



Why is VMD Non-Causal? The Global Optimization Trap

ADMM Optimization: Iterative whole-window solving. It optimizes the sum of all modes across all time steps simultaneously.

$$\min \sum \left\| \partial_t [u_k(t) \dots] e^{-j\omega_k t} \right\|_2^2$$

RFFT (Fast Fourier Transform): Global Frequency Scan. FFT assumes the signal is a complete, cyclic whole.

$$\sum u_k = f(t)$$

Reconstruction Constraint: Forces the algorithm to balance past and future continuously.

VMD doesn't filter sequentially step-by-step;
it scans the entire 240-day window simultaneously.

The 3 Fatal Dangers for Forecasting



Look-Ahead Bias (Data Leakage)

Future target data bleeds into training IMFs.
Yields 100% fake accuracy in training, total failure in live trading.



Boundary Effects (Gibbs Phenomenon)

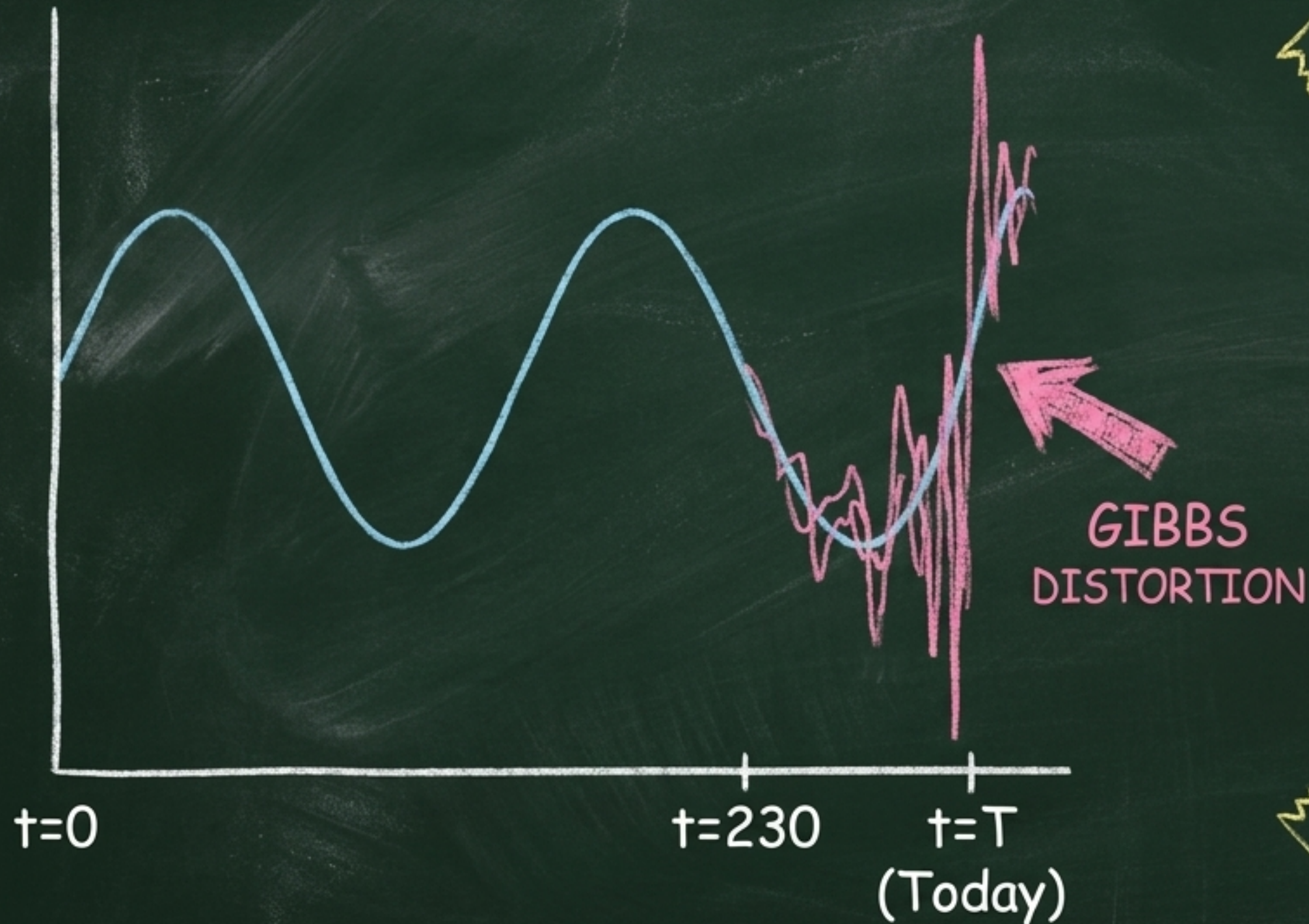
FFT assumes infinite looping.
Causes severe numerical distortion at the exact edges of the data.



Temporal Inconsistency

As the sliding window moves, historical IMF values constantly shift and rewrite themselves.
The AI learns to hit a moving target.

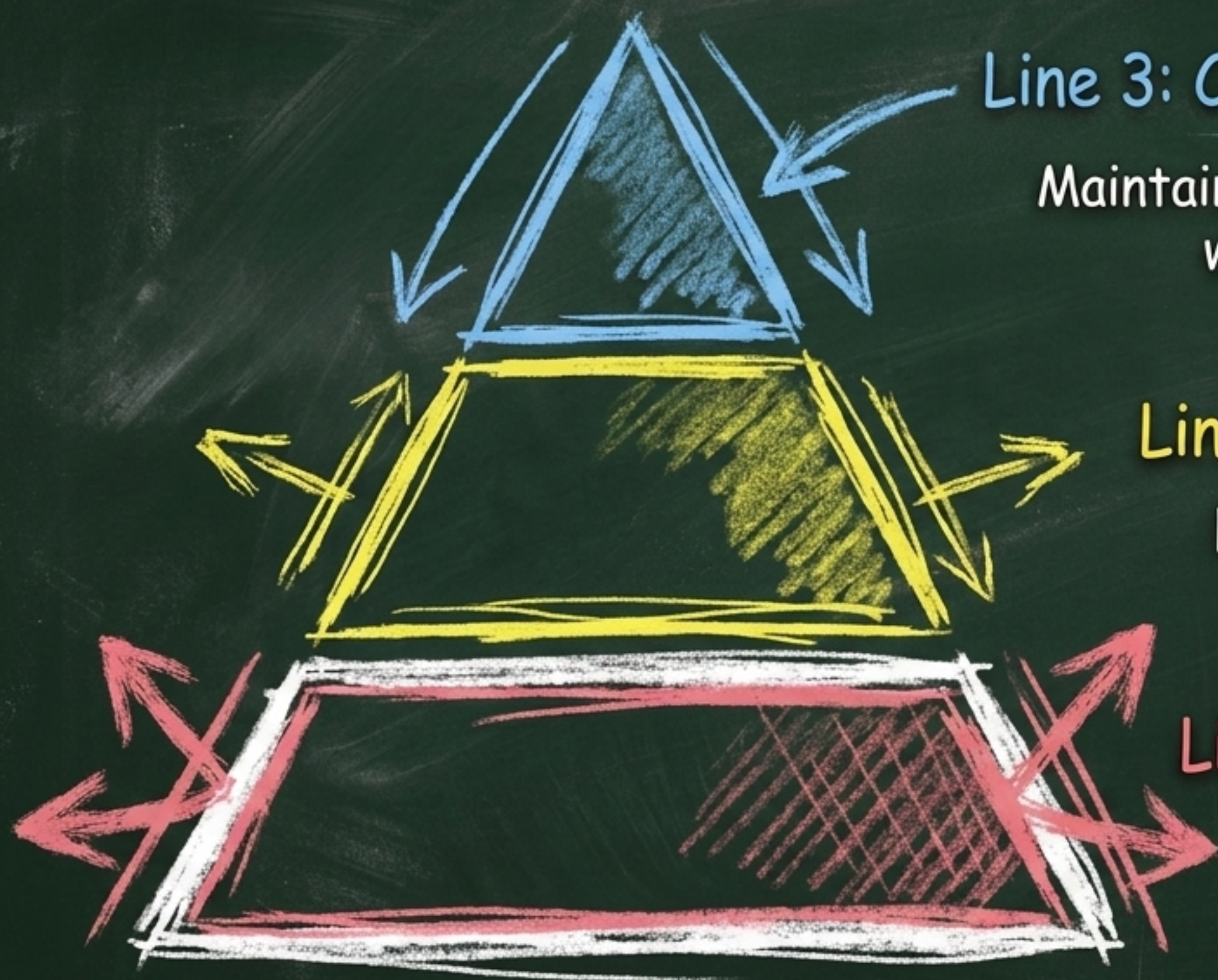
The Critical Vulnerability: The Edge of $t=T$



In forecasting, the most important data point is the absolute edge of the window: $t=T$ (Today). It is the launchpad for tomorrow.

Because of the Gibbs Phenomenon, VMD experiences violent frequency snapping at the boundaries. This corrupts the exact phase and amplitude the SDE engine needs to start its simulation.

PhaseSync SDE: The 3-Line Defense



Line 3: Causal State-Space Warm Starts

Maintains temporal consistency across sliding windows. Prevents mode-hopping.

Line 2: Mirror Padding & Cropping

Flushes Gibbs boundary artifacts to the outside of the computational frame.

Line 1: Batch Sliding Window Isolation

Strict physical firewall against future data leakage into the training graph.

Defense 1: Strict Temporal Isolation

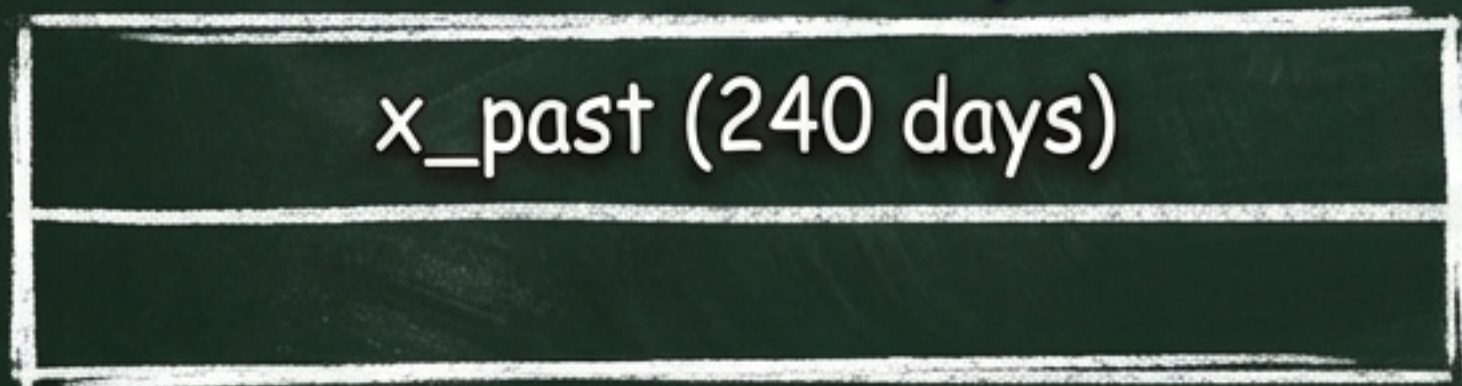
Naive Approach (NEVER DO THIS)



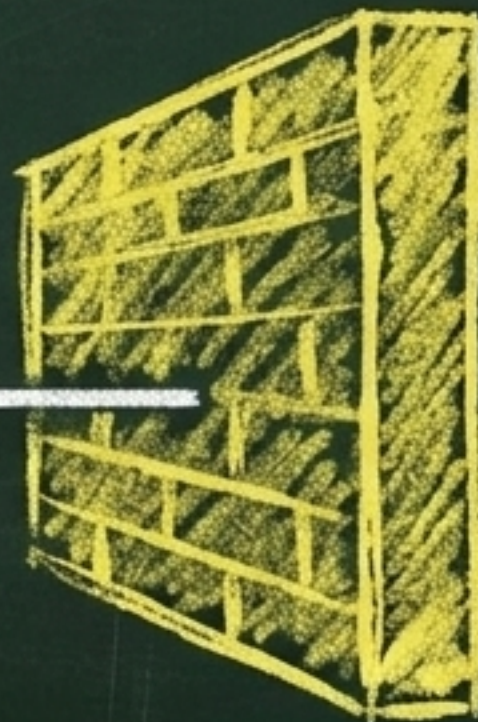
[Train (Past) + Test (Future)]

VMD calculates over the whole thing

BatchVMDLayer



VMD operates ONLY inside here

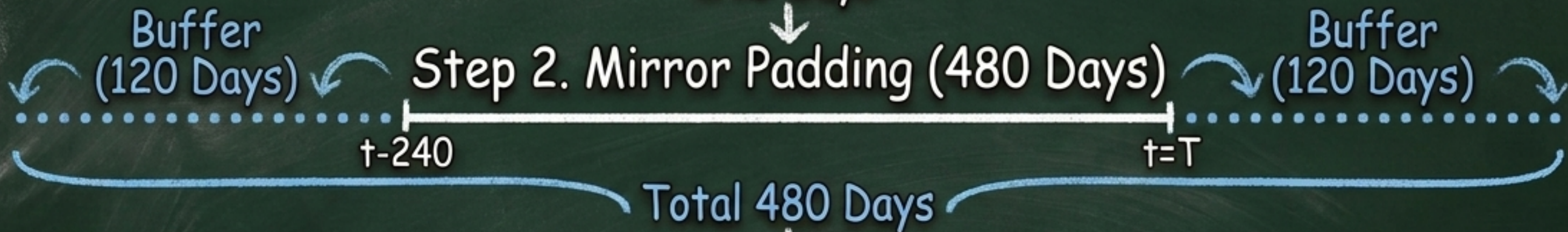
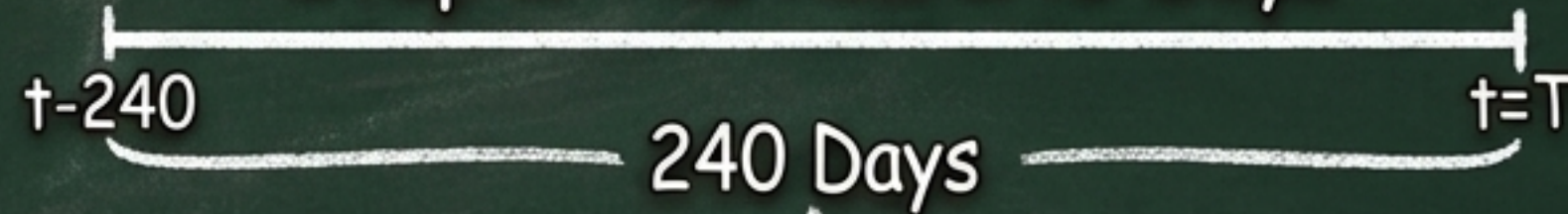


100% Future Data Blocked
outside the computational graph.

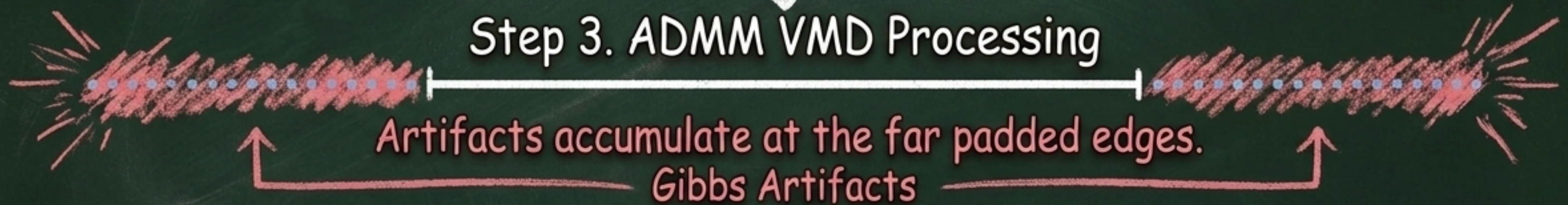
y_true (Future 24 days)

Defense 2: Mirror Padding & Surgical Cropping

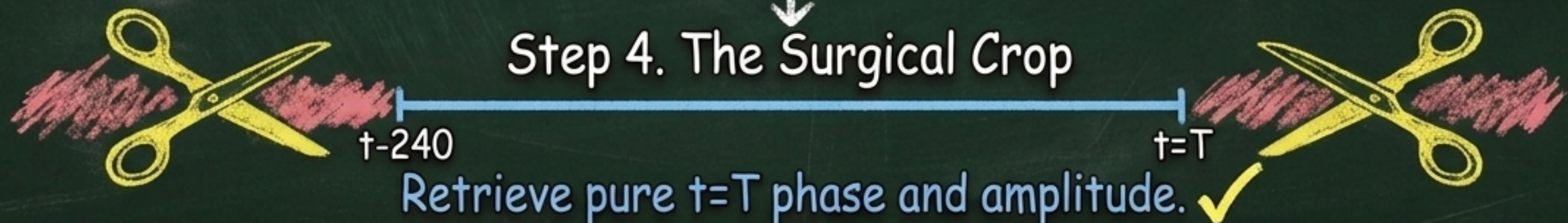
Step 1. The Raw 240 Days



Step 3. ADMM VMD Processing



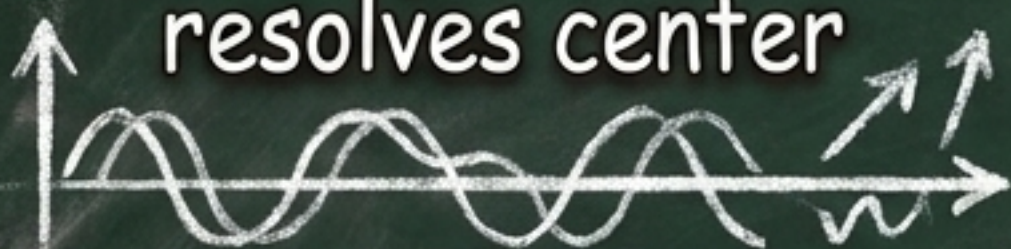
Step 4. The Surgical Crop



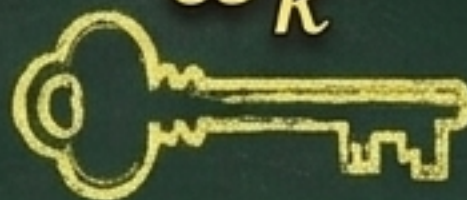
Defense 3: State-Space Warm Starts

Window_T (Today)

VMD optimization
resolves center



ω_k



Window_{T+1} (Tomorrow)

ADMM optimization
receives the key
as its Prior.

Instead of solving VMD from scratch every day (which causes mode-hopping and instability), we pass the center frequencies (ω_k) from yesterday as the Prior.

- ✓ Forces temporal consistency
- ✓ Tracks regime shifts smoothly without rewriting history

Bonus Defense: Restoring Spatial Independence

CNN Encoder (The Problem)



Noise is 100% correlated.

Breaks the conditional independence assumption of the KvM Closed-form solution.

Creates Spurious Coupling.

VMD Pre-processing (The Solution)

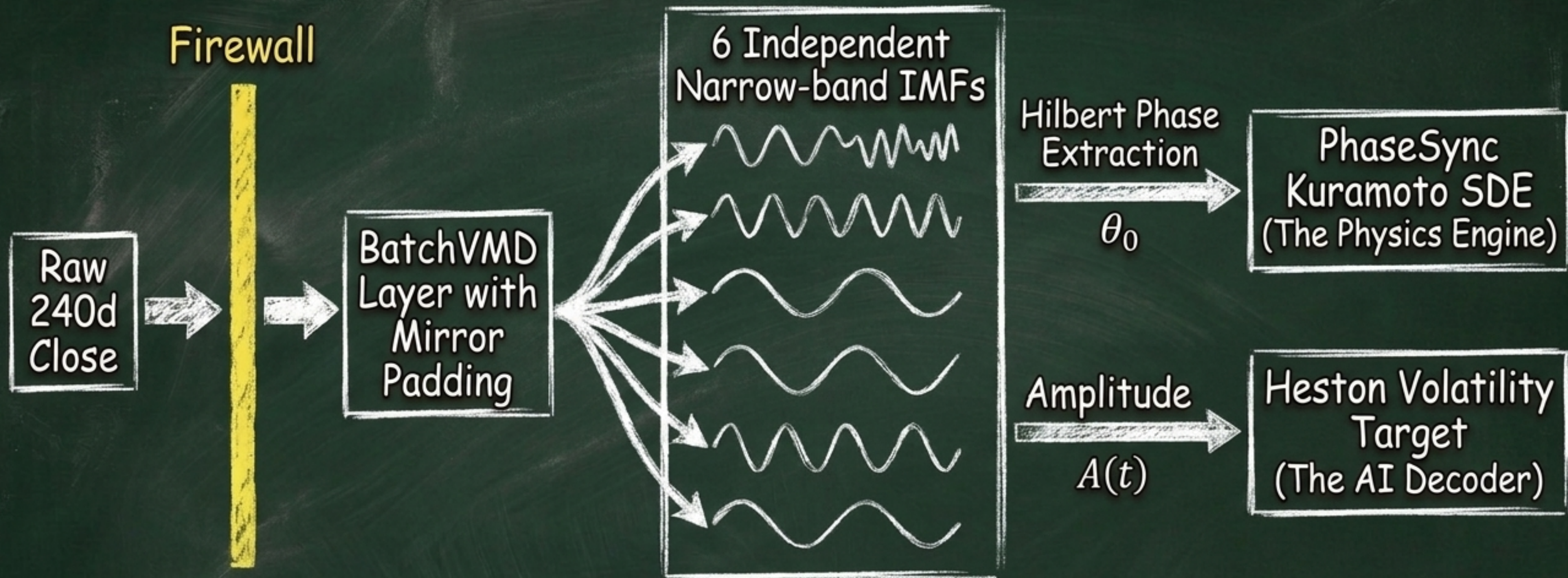


IMFs are mathematically orthogonal (non-overlapping frequencies).

Distributing 6 IMFs to 48 oscillators restores Spatial Conditional Independence.

The Closed-form MLE calculates true, physical Lead-Lag relationships.

The Causal VMD-SDE Pipeline



By sanitizing VMD, we feed perfectly causal, independent frequency bands into the 2nd-Order SDE, eliminating spurious coupling and look-ahead bias.

Conclusion: VMD is a Scalpel, Not a Crystal Ball

Standard VMD:

- ⚠ Non-Causal Global Optimization
- ⚠ Suffers from Look-Ahead Bias
- ⚠ Severe Edge Artifacts at $t=T$
- ⚠ Prone to Mode Hopping & Temporal Inconsistency

PhaseSync Causal VMD:

- ✓ Strict Batch Sliding Isolation
- ✓ 100% Future Data Walled Off
- ✓ Mirror-Cropping Flushes Artifacts
- ✓ State-Space Warm Starts & Spatial Independence

VMD out of the box is non-causal and dangerous for forecasting. Through strict isolation, padding, and state-tracking, PhaseSync transforms it into a perfectly causal, high-precision scalpel for financial SciML.